

KIRILL SKOBELEV

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EDUCATION

Northwestern University

Ph.D. in Computer Science. Advisor: [Jessica Hullman](#)

Evanston, IL
Sep. 2026 – June 2031 (expected)

New Economic School & Higher School of Economics

B.A. in Economics, *summa cum laude*, GPA: 9.38/10.0

Moscow, Russia
Sep. 2020 – June 2024

EXPERIENCE

University of Chicago, Center for Applied AI

Research Professional

Chicago, US
Aug. 2024 – Present

- (ongoing) RA to Prof. [X.Y. Han](#): Evaluated 19 open-weight and 5 proprietary frontier VLMs on surgical tool detection across 67,634 neurosurgery frames and a public cholecystectomy benchmark. Showed zero-shot VLMs fail to surpass a majority-class baseline regardless of scale (2B–235B params), fine-tuning hits a generalization ceiling under distribution shift, and a 26M-parameter YOLO model outperforms all VLMs with $1,000\times$ fewer parameters. [[paper](#)]
- (ongoing) RA to Prof. [Rimmy Tomy](#): Examined whether U.S. financial regulators' climate risk guidance reflects implicit policymaking by analyzing public comment influence on draft-to-final revisions using LLMs. [[paper draft](#)] [[code](#)]
- With [Eric Fithian](#). Co-developed an open-source Python toolkit addressing reproducibility and scalability challenges in LLM-based data extraction. [[paper](#)] [[code](#)]
- RA to Prof. [Kilian Huber](#): Extracted commodity price expectations from over 400,000 investor call transcripts, achieving 90% precision against human labels. Designed a token-efficient method for compressing few-shot examples. [[paper](#)] [[code](#)]
- RA to [Daniel Albohn](#) and Prof. [Alex Todorov](#): Compiled and cleaned a dataset of over 2,500,000 images of food items. Trained a StyleGAN model to generate synthetic images of food items to study social perceptions of food.
- RA to Prof. [Sanjog Misra](#) and Prof. [Max Farrell](#): Co-developed a replication package for parameter inference in economic models, using neural networks as parameters within structural frameworks. [[paper](#)] [[code](#)]

Banco Plata

Data Scientist at Credit Portfolios

Mexico City, Mexico
Nov. 2023 – June 2024

- Built and integrated ML forecasts into a customer-level NPV model for portfolio decisions.

WORKING PAPERS

Skobelev, Kirill, Eric Fithian, Yegor Baranovski, Jack Cook, Sandeep Angara, Shauna Otto, Zhuang-Fang Yi, John Zhu, Daniel A. Donoho, X.Y. Han, Neeraj Mainkar, and Margaux Masson-Forsythe. *A Comparative Study in Surgical AI: Datasets, Foundation Models, and Barriers to Med-AGI*. 2026. arXiv preprint. [[arXiv:2603.27341](#)]

Eric Fithian and **Skobelev, Kirill**. *DELM: a Python toolkit for Data Extraction with Language Models*. 2025. arXiv preprint. [[arXiv:2509.20617](#)] [[code](#)]

Skobelev, Kirill and Rimmy E. Tomy. *Bank Regulators and Climate Action: Evidence from Supervisory Guidance*. 2025. Preprint forthcoming. [[pdf draft](#)] [[code](#)]

Skobelev, Kirill. *Statistical Learning Meets Analyst Forecasts of Corporate Earnings*. 2024. Undergraduate Thesis. [[pdf](#)]

Skobelev, Kirill. *Bitcoin Production Cost: Demystifying the Mining Industry*. 2021. SSRN preprint. [[SSRN:4005400](#)]

HONORS AND AWARDS

Graduated summa cum laude, New Economic School	2024
Fellow of Lev Lyubimov scholarship for outstanding academic achievement	2022
Winner of the Student Research Paper Competition held by the HSE University	2021
Fellow of LaToken Decentralized Finance scholarship	2021
Winner of the final stage of the All-Russian School Olympiad in Economics [1 in 2,000]	2020
Graduated with highest honors from EMSCH school at Lomonosov Moscow State University	2020

VOLUNTEERING AND OTHER

Undergraduate thesis defense committee member (AI+Economics), New Economic School	2026
Undergraduate thesis defense committee member (AI+Economics), New Economic School	2025
Member of the jury for the Moscow School Competition in Economics	2022
Teaching Assistant, Introduction to Economics at the New Economic School, rating: 4.73/5.0	2021
Authored tasks for the regional stage of the All-Russian School Olympiad in Economics	2021

SKILLS

Coding: Python (pandas, numpy, torch, xgboost, transformers, optuna, sklearn, bs4), git, SQL, R (tidyverse), slurm, AWS.

Languages: English (fluent), Russian (fluent), Spanish (conversational).